

Digital Asset Cryptography

Part 4 - Ethereum & L2s: Protocol Quantum Roadmap

Compiled by the Spaced Research Ledger

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Ethereum is the only major chain with a staffed team, a published roadmap and running test networks devoted to post-quantum migration. The work began in earnest in January 2026 and Vitalik Buterin has publicly confirmed the timeline. The strategy differs from Bitcoin's proposals in Part 3 in a basic way. Instead of a deadline that sunsets old signatures, Ethereum plans to let each account switch to new cryptography on its own schedule.

The hard technical problem is not signing but aggregation. Ethereum's consensus depends on compressing hundreds of thousands of validator signatures into one, and no post-quantum scheme does that natively. Most of the engineering below exists to answer that one gap.

Section 1 covers the roadmap and the team behind it. Section 2 covers the four cryptographic systems that need replacing. Section 3 covers the aggregation problem and the lean component stack. Section 4 covers milestones and the 2029 target. Section 5 covers the account-abstraction change nearest to shipping, and Section 6 the open research.

1. The Roadmap and Who Owns It

Post-quantum migration on Ethereum is an assigned institutional project with named teams, not a mailing-list debate.

Current State

The Ethereum Foundation established its Post-Quantum Security team in January 2026, and the roadmap, called the Strawmap, was published at strawmap.org after a workshop that month.[1] Vitalik Buterin publicly confirmed the roadmap and its timeline in late February 2026.[1] The work sits with the Post-Quantum team inside the Foundation's Protocol cluster, with a parallel Protocol Snarkification team under Alex Hicks focused on formal verification.[2]

The work spans three protocol layers. The execution layer gets account abstraction, explicitly without a disruptive flag day. The consensus layer replaces its BLS signatures with post-quantum alternatives. The data layer secures data availability with post-quantum cryptography.[2] The plan unfolds in three phases: readiness and infrastructure, gradual adoption, and protocol-level consolidation.[2]

2. What Needs Replacing

Four separate cryptographic systems inside Ethereum are quantum-vulnerable, and each gets a different replacement.

Current State

The official roadmap identifies four systems needing replacement.[3] Consensus-layer BLS signatures carry validator votes and rely on elliptic-curve pairings. KZG commitments provide data availability for rollups and are also pairing-based. ECDSA account signatures on the secp256k1 curve expose their public keys once a transaction is sent. And application-layer zero-knowledge proofs are built on pairing-based SNARKs.[3]

The planned replacements follow the same order. BLS gets leanXMSS plus leanVM, and KZG gets STARK-based or lattice-based commitment schemes.[3] Accounts reach post-quantum schemes through account abstraction.[3] The proofs migrate through natural ecosystem adoption of STARKs, which are already quantum-resistant.[3]

3. The Aggregation Problem and the Lean Stack

Quantum-safe signatures exist. A quantum-safe way to combine half a million of them into one small proof did not, and building it is most of the project.

Current State

The hard part is not quantum-safe signing but quantum-safe aggregation. BLS can combine hundreds of thousands of validator signatures into one, and no post-quantum scheme has an obvious equivalent, while post-quantum signatures are far larger to begin with.[4] The size gap is roughly 3,000 bytes for a hash-based validator signature against 96 bytes for BLS. The answer to that gap is leanVM, a minimal virtual machine acting as a proof-based aggregation engine, compressing by about 250 times.[3] The signature scheme itself, leanXMSS, is quantum-safe because it depends only on hash-function security, which quantum computers weaken but do not break.[3]

The component set has five named pieces. leanXMSS provides hash-based validator signatures and leanVM the aggregation engine.[2] leanSig is a Rust implementation of the signing scheme.[2] leanSpec is an executable Python specification in use by about ten client teams, and leanMultisig implements the aggregation.[2]

Status varies by piece. leanSig is integrated, with verification running at 39% of its target performance. leanMultisig is in testing at about 97% of target efficiency on recent Apple hardware. leanVM remains in development.[5]

Test networks are running now. The sequence began with pq-devnet-0 in October 2025 and reached pq-devnet-4, active as of March 2026, which performs recursive post-quantum signature aggregation using leanVM. A fifth devnet is planned to move aggregation proofs to the block level.[5] Weekly interoperability devnets run with more than ten client teams.[2]

4. Milestones and the 2029 Question

The roadmap targets protocol-level completion around 2029. That is not the same as Ethereum being post-quantum in 2029, and the roadmap says so.

Current State

Reported scope of the migration includes six quantum-resistant signature schemes and 13 virtual-machine precompiles, with full activation planned before 2030.[1] The structured milestones target completion of core post-quantum infrastructure by about 2029. Milestone I is a post-quantum key registry for validators, and Milestone J is signature verification precompiles.[3] Milestone L is consensus attestations through leanVM, and Milestone M is full aggregation plus safe blob commitments.[3] The roadmap page gives rough markers rather than fixed dates for these.[2]

The team's own words bound the claim. Protocol upgrades at the base layer could be completed by 2029, with full execution-layer migration taking additional years beyond that.[2] The nearest-term carrier is the Hegotá fork, scheduled for the second half of 2026, which implements EIP-8141.[3]

5. EIP-8141 and Signature Agility

The account-abstraction change closest to shipping is also the escape hatch from vulnerable keys. Ethereum's bet is voluntary migration, account by account.

Recent Developments

EIP-8141 is targeted for the Hegotá fork as Considered for Inclusion, and was formally added to the fork's coordinating document in a change merged on 30 April 2026.[6] A dedicated tracking site, eip8141.io, documents the divergence between the original January specification and the evolved current one.[6] It also lays out a seven-stage post-quantum roadmap, running from decoupling ECDSA to private base-layer settlement.[6] A late-August research proposal would extend the ecrecover opcode, the instruction contracts use to check signatures.[7] It would recognise post-quantum signatures from frame transactions through reserved sentinel values, without disabling the opcode.[7]

Current State

EIP-8141, titled Frame Transaction, is a draft core-protocol change created on 29 January 2026. Its ten authors include Vitalik Buterin and several account-abstraction researchers.[8] It introduces a transaction type that abstracts validation, execution and gas payment through a frame system. Transactions decompose into contract calls that determine their own validity and payment, rather than relying on fixed ECDSA keys.[8]

The stated purpose is explicit. The proposal provides a native off-ramp from today's elliptic-curve authentication to post-quantum systems, and unlinking accounts from specific keys enables native key rotation.[8] The document itself does not name the fork that will carry it, even though the roadmap page assigns it to Hegotá.[8] The wider strategy is signature agility through account abstraction rather than a protocol-wide forced migration. Individual accounts adopt post-quantum signatures independently, a deliberate architectural contrast with the deadline-and-sunset approach proposed for Bitcoin.[3]

6. Open Research

The parts of the problem without settled answers live on the research forum, and they are aggregation-shaped.

Recent Developments

A June 2026 research post explores the design space for a post-quantum public key registry for Ethereum validators. It proposes a phased rollout, registry fork first and signature switch later, using leanVM-style proof systems for aggregation.[9]

Current State

Post-quantum attestation aggregation remains an open research area with live proposals, including separator-based participation commitments and a folding-based approach to signature aggregation.[10] The academic side is working the same problem. A 2026 cryptology preprint, titled Transient Quantum Resistance, with Application to Ethereum Consensus, addresses the consensus layer directly.[11]

About This Report

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